

A Beginner's Practical Guide to CI/CD Pipelines

CI/CD is the backbone of modern software engineering that automates how code goes from your laptop to the cloud. Mastering these fundamentals saves you from manual deployment stress and helps you ship reliable apps safely.

What you'll learn:

1. Version Control with Git
2. Automating Checks with GitHub Actions
3. Packaging Apps with Docker
4. Automated Deployment with CD

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1. Version Control with Git

Before you can automate anything, you need to track your code changes safely. Git allows you to work on features without breaking the main app and lets you travel back in time if something goes wrong. Every CI/CD pipeline starts with a simple git push command.

```
$git add .  
git commit -m "Add user profile page"  
git push origin main
```

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2. Automating Checks with GitHub Actions

Instead of manually testing your code every time you save, CI tools run tests for you automatically. When you push your code to GitHub, it spins up a tiny virtual computer to run your build and test scripts before letting anyone merge the code.

```
$name: Run Tests
on: [push]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - run: npm install
      - run: npm test
```

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3. Packaging Apps with Docker

Have you ever heard the phrase 'it worked on my machine'? Docker solves this by packing your app and everything it needs into a neat little box called a container. This ensures your app runs the exact same way on your laptop and in production.

```
$docker build -t my-web-app .  
docker run -p 8080:8080 my-web-app
```

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4. Automated Deployment with CD

Continuous Deployment takes your tested code package and automatically uploads it to your cloud server. This removes human error from the release process and gets your new features into the hands of users in seconds.

```
$aws ecs update-service --cluster production --service web-app --force-new-deployment
```


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Key Takeaways

- + Always test your code locally before pushing to GitHub.
- + Keep your CI build times fast by caching dependencies.
- + Use environment variables for passwords and secret keys.
- + Add automated tests incrementally as your project grows.
- + Read your pipeline error logs carefully when builds fail.

Watch Out For

- ! Putting passwords and API keys directly inside your code files.
- ! Trying to automate 100% of your deployment on day one.
- ! Ignoring failed test alerts instead of fixing the root cause.

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